



EEP-AGAPP - Agentic AI Applications Design Doc

Basic information

Course code name: MLU-EEP-AGAI-M2

Course name: Agentic AI Applications: from Concept to Code

Projected full duration: 4 hours (Can vary depending on modularization based on students' personas)

Short course description

This lesson provides a technical deep dive into autonomous AI system development for faculties. The lesson covers methodologies (Chain of Thought, ReAct, Plan & Execute, Reflexion) and essential architectural components including plan generation, function calling, memory management, error handling, and human-in-the-loop collaboration. Faculties will analyze framework options like Strands, and LangGraph, with special emphasis on safety considerations for responsible development. The lesson includes practical demonstrations and activities that will reinforce the theoretical concepts for teaching purposes.

Long course description

As AI continues to evolve, there's a growing need for faculties to understand the power of AI systems that can reason, act, and make decisions autonomously. Traditional approaches to AI implementation often fall short when dealing with complex, multi-step tasks that require reasoning, planning, and adaptation.

This comprehensive lesson provides a technical exploration of agentic AI application development, equipping faculties with the knowledge and practical skills needed to learn and practice how to design, implement, and deploy custom AI agents. Faculties will master core prompting methodologies that enable continuous improvement through self-reflection.

The lesson covers essential architectural components crucial for robust autonomous systems. These include plan generation, function calling for interacting with external tools and APIs, memory management, error handling, and human-in-the-loop collaboration. Faculties will explore popular frameworks such as Strands agents, and LangGraph.

Throughout the lesson, special emphasis is placed on safety considerations and responsible AI development practices. The lesson features hands-on demonstrations and activities that reinforce theoretical concepts, preparing faculties to effectively learn about these advanced topics.

How challenging is this course?

Difficulty level: 200

What category does this course fall into?

Agentic AI

Profiling information

Learners

Target Audience: The audience for the EEP program is faculty in higher education institutions as part of MLU Education Enablement Program, who want to integrate Agentic AI content into their skills and knowledge, preparing for teaching this content.

To make the current **MLU AGAPP – Agentic AI Applications** content accessible to community college faculties, we must account for their diverse backgrounds—across both code-based and code-free programs. By creating tailored user stories for these two distinct faculty personas, the content can be adapted to address their unique learning needs and teaching goals.

These persona-based user stories will also support content modularization, giving faculty members the flexibility to choose modules that are most relevant to their students' personas and educational backgrounds.

The following user stories, based on these two personas and this learning material, define what each learner should be able to accomplish upon course completion:

code-based faculty

This faculty prepares and teaches courses for degrees in hard sciences, engineering, mathematics, or related technical disciplines such as physics, computer science, or mechanical engineering. Their software development experience ranges from beginner to intermediate—some may have completed programming courses or personal projects, while others are just starting. They're comfortable with fundamental coding concepts like loops, functions, and conditionals, as well as data structures including arrays, lists, and trees.

code-free faculty

This faculty prepares and teaches courses for degrees in business, healthcare administration, liberal arts, or similar non-technical disciplines. They possess strong analytical and critical thinking abilities and recognize technology's influence on their field, yet their technical knowledge remains limited to everyday consumer applications. Relevant examples include soft science and business courses such as social sciences or health sciences. While technical subjects may seem daunting, they're eager to grasp AI technology fundamentals and practical applications.

Content Modularization

Lecture activities

All MLU EEP course content is modularized, allowing faculty members the flexibility to choose modules that are most relevant to their educational backgrounds.

Throughout our course's slide deck, we use two different icons to identify the proposed target persona relate to the content:

 *code-based Course Icon*

 *code-free Course Icon*

Both lecture content, supported by slide decks, and lab hands-on material are identified by these icons.

Usage example

If you your background is code-free, look for all slides that have this icon: 

otherwise, look for all slides that have this icon: 

Hands-on activities

Hands-on activities also are modularized considering three different personas, adding two categories to the technical personas:

 *code-free faculty*

These hands-on activities aim to give experience creating AI solutions using Amazon Quick. This no-code environment is ideal for faculties without programming background, enabling them to build sophisticated AI solutions effortlessly.

  *Engineering focused faculty with software engineering path*

These hands-on activities are suited for personas following a software engineering path but also can be used for general code-based faculty personas. The activities use Amazon Kiro coding assistant IDE to practice Agentic AI concepts by implementing Bedrock Agentic features. This environment is ideal for faculties with some coding experience, allowing them to explore how AI-powered coding assistants can accelerate development and help build robust AI systems efficiently.



These hands-on activities are suited for personas following a science path but also can be used for general code-based faculty personas. The activities use the Amazon SageMaker Studio environment to practice Agentic AI concepts by implementing Bedrock Agentic features. This environment provides an ideal workspace for data scientists experimenting on different AI models and architectures to address business challenges, without software and hardware infrastructure distractions.

User Stories

ID	User Stories, what do you want the people to be able to do when they leave the class? As a ___, I want ___ so that ___
Code-based faculty 	As a code-based faculty , I want to understand how to explain fundamental reasoning techniques like Chain of Thought (CoT), ReAct, and Plan & Execute so that I can implement these in my AI projects.
	As a code-based faculty , I want to understand how reasoning models solve complex reasoning tasks so that I can use these concepts effectively in advanced programming courses.
	As a code-based faculty , I want to differentiate between various reasoning methodologies and their appropriate applications so that I can make informed design decisions in my agent development projects.

	As a code-based faculty , I want to understand how to simplify complex agentic concepts so that I can learn these foundational topics without being overwhelmed.
	As a code-based faculty , I want to demonstrate practical examples of function calling and tool selection, understanding how agents interact with external systems.
Code-free faculty 	As a code-free faculty , I want to discuss human-in-the-loop collaboration frameworks and understand how to design trustworthy AI systems in professional contexts.
	As a code-free faculty , I want to explain agentic architectures at a conceptual level, understanding strategic implications for business applications.
	As a code-free faculty , I want to explain agent capabilities and limitations in simple terms and learn how to make informed decisions about AI adoption in my future career.
	As a code-free faculty , I want to learn how AI agent tools can enhance and streamline my personal and academic tasks.
All personas 	As a faculty , I want to understand safety risks specific to autonomous agent systems so that I can learn responsible AI development practices, regardless of my technical background.
	As a faculty , I want to effectively create and use AI tools to support my academic activities, regardless of my technical background.

Prerequisites

Prerequisite	Required or Optional?	Why is this a prerequisite?
Completion of EEP-AGAI - Module 1 - Agent Essentials course	Required 	This lesson continues the discussion from AGESS which lays down the foundation for agentic AI concepts
Working knowledge of Python and Bedrock API usage for LLM invocation	Required 	For hands-on demonstrations, working knowledge of Python and LLM calling via Bedrock is necessary for model invocation, parameter configuration, and response handling
Understanding of Gen AI fundamentals	Required 	Need to understand how LLMs process and generate text; Knowledge of token limits and their implications; Understanding of model capabilities and limitations; Ability to choose appropriate models for specific tasks; Understanding temperature and other parameter settings
Understanding of Basic Prompting techniques like ReAct	Required 	Crucial for grounding agent responses in factual information; Helps mitigate hallucinations and improve accuracy; Provides framework for structured reasoning and action planning; Enables step-by-step problem solving and decision making; Crucial for creating agents that can break down complex tasks

Learner Outcomes

ID	Action Verb	(Finish the sentence: After completing this lesson, faculty are prepared to...)	How will faculty demonstrate they have learned this?
1	Explain	fundamental reasoning techniques like Chain of Thought (CoT), ReAct, and Plan & Execute in agentic AI systems	Lecture adaptation, Discussion facilitation
2	Discuss	how reasoning models solve complex reasoning tasks	Lecture delivery, Case examples
3	Differentiate	between various reasoning methodologies and their appropriate applications in agent design	Lecture content, Comparative analysis activities
4	Discuss	how Reflexion improves agent performance through multiple reasoning pathways and reward signals	Lecture delivery, Demonstration activities
5	Apply	function calling and tool selection techniques	Lab demonstration, lab guided activities
6	Solve	common execution failures using autonomous error handling and unblocking strategies	Demonstration, Problem-solving activities

7	Apply	short-term and long-term memory management concepts	Lab demonstration, lab guided activities
8	Discuss	human-in-the-loop collaboration frameworks for enhancing agent trustworthiness in various contexts	Lecture delivery, Discussion facilitation
9	Differentiate	established agentic frameworks based on their core characteristics and use cases	Lecture content, Framework comparison activities
10	Discuss/ Apply	safety risks specific to autonomous agent systems including compounding errors and unbounded solution spaces	Lecture delivery, Discussion activities

Completion Criteria

Completion criteria will be defined by faculty members using this content. Below are some suggestions:

Criteria	Details
Quiz/Evaluation	Questions covering the learning outcomes like LLM Reasoning, tool use, plan generation, error handling, memory management, and safety considerations
Participation	Active engagement in discussions and demonstrations about reasoning methodologies, frameworks, and responsible AI practices

Teaching Application	Successful adaptation of technical concepts for different faculty personas and skill levels by building an AI agentic app.
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Detailed topic list

Flexibility is a key driver for implementing MLU EEP content. Faculty members will adapt the content according to their own instructional plans. The table below provides general suggestions for topic coverage aligned with the learning objectives.

Activity #	Name	Topic covered	Additional Details	Learner Outcome:
Section 1	Reasoning and planning	Reasoning in LLMs	This text covers LLM reasoning techniques: Chain-of-Thought (CoT) explores implicit reasoning and training methodologies; ReAct demonstrates interleaved reasoning-action for complex tasks; Reflexion uses reward-guided multiple reasoning pathways to improve agent performance through feedback mechanisms; Adaptive Graph of Thoughts (AGoT) , moving beyond fixed-step methods like Chain of Thought and Tree of Thoughts through recursive decomposition of complex queries.	LO1: Explain fundamental reasoning techniques like Chain of Thought (CoT), ReAct, and Plan & Execute in agentic AI systems LO3: Differentiate between various reasoning methodologies and their appropriate applications in agent design LO4: Discuss how Reflexion improves agent performance through

				multiple reasoning pathways and reward signals
		Reasoning and Planning Techniques	This module examines standard agentic framework execution steps: detailed plan generation (plus plan-as-you-go alternatives), tool selection via function calling, agentic RAG foundations, reflection and plan refinement based on outcomes, and autonomous error handling for unblocking failures.	LO5: Apply function calling and tool selection techniques LO10: Solve common execution failures using autonomous error handling and unblocking strategies
		Unified agentic multimodal processing	This module introduces the emergence of unified multimodal processing. We will cover its ability to process images, text, audio, and video simultaneously while demonstrating strong proficiency in both speech and image generation.	LO2: how reasoning models solve complex reasoning tasks
		Reasoning Models	This module introduces reasoning models, their key characteristics, and why they are important to solve complex problems.	LO2: how reasoning models solve complex reasoning tasks
Section 2		Building your First Agent	Explore the lifecycle of reasoning model implementation, covering also the balance between complexity and high precision.	LO3: Differentiate between various reasoning methodologies and their

	Building and Customizing Agents			appropriate applications in agent design
		Lab 2.1 Compare Agents vs No-Agents Lab	This hands-on lab demonstrates the fundamental distinctions between agentic systems and conventional LLM-based applications with tools. Faculties will solve an identical task using both methodologies, clearly revealing how agentic frameworks operate autonomously compared to the step-by-step procedural approach of traditional implementations. Through this direct comparison, the unique capabilities and advantages of agentic systems become evident.	LO5: Apply function calling and tool selection techniques LO6: common execution failures using autonomous error handling and unblocking strategies
		Session and memory	This module explores how agents maintain session context and overcome statelessness through attribute management. It covers dual memory systems (short-term and long-term) enabling coherent, personalized responses. Topics include memory management techniques, dynamic updating strategies for cost reduction, innovations like mem0 with practical applications, episodic memory systems, neurobiologically inspired memory models, and dynamic memory management strategies.	LO7: Apply short-term and long-term memory management in agentic systems
		Controlling and Validating Flows	This module covers agent flows customization, how to differentiate between execution and moderation	LO3: Differentiate between various reasoning

			properly, the importance of session and agent state, as well as the inclusion of human-in-the-loop validation.	methodologies and their appropriate applications in agent design LO10: human-in-the-loop collaboration frameworks for enhancing agent trustworthiness in various contexts
		Model Context Protocol (MCP)	This module covers the introduction of model context protocol (MCP), its components, and connection lifecycle.	LO5: Apply function calling and tool selection techniques
		Agentic frameworks	This module builds upon previous conceptual discussions by examining established agentic frameworks, strands, LangGraph, and others. Rather than delving into implementation specifics or code examples, the focus remains on the core characteristics of each framework. We analyze framework selection criteria, identifying optimal use cases while weighing the advantages and disadvantages for various common agentic applications previously discussed. The comparative analysis covers essential framework attributes including underlying execution processes, managed versus self-hosted service models, information	LO9: Differentiate established agentic frameworks based on their core characteristics and use cases

			flow architectures, memory management approaches, human-in-the-loop integration techniques, agentic protocol support, and documented limitations. Additional framework attributes will be incorporated during course development to ensure comprehensive coverage. We also present basic architecture characteristics for IoT integration, addressing business transformation opportunities using edge computing, distributed agent coordination, and real-time sensor data processing.	
		Lab 2.2 LLM reasoning and planning with memory Lab	Demonstrate how session attributes such as memory enables the agent to execute complex tasks effectively	LO5: Apply function calling and tool selection techniques LO10: Discuss safety risks specific to autonomous agent systems including compounding errors and unbounded solution spaces
		Agentic AI for Software Development	This module covers key concepts and functionalities of an agentic AI coding assistant platform to support	LO2: how reasoning models solve complex reasoning tasks

			building software systems. We will present the key functionalities of Amazon Kiro, used in the hands-on lab.	
		Lab 2.3 Building AI agents without coding	Hands-on lab where faculties explore how AI agents autonomously plan and execute multi-step tasks to support their research and study activities. Using Amazon Quick—a no-code platform—faculties will improve the AI assistant created in lesson 1, adding a research plan to the app's features.	LO2: how reasoning models solve complex reasoning tasks
		Lab 2.4 Creating a simple LLM app using AI coding assistant	Hands-on lab where faculties explore how to create AI agents exploring Kiro IDE AI coding assistant to create an app to help their research and study activities.	LO2: how reasoning models solve complex reasoning tasks LO5: Apply function calling and tool selection techniques
Section 3	Building Responsible AI	Increasing autonomy of agents	This module addresses safety and responsibility in agentic AI systems, emphasizing robust evaluation and safety measures. Key risks include autonomous execution, accuracy degradation, unbounded solution spaces, dual LLM roles, and model limitations. It incorporates standard evaluation methodologies and	LO10: Discuss safety risks specific to autonomous agent systems including compounding errors and unbounded solution spaces

			observability tools for developing responsible, accountable agentic systems.	
		Risks and attack surfaces for AI agents	This module presents and discusses different kinds of attacks that can target agents.	LO10: Discuss safety risks specific to autonomous agent systems including compounding errors and unbounded solution spaces
		Risk mitigation strategies	This module presents the risk mitigation framework and discusses how they can be used to mitigate different threats.	LO10: Discuss safety risks specific to autonomous agent systems including compounding errors and unbounded solution spaces
Session 5	Evaluating AI Agents	Evaluation and observability	This module explores the complexity behind evaluating complex agent environments, presenting different evaluation metrics.	LO9: Differentiate established agentic frameworks based on their core characteristics and use cases
		Evaluation and observability frameworks	This module presents evaluation and observability frameworks that help build a more robust agentic solution.	LO6: common execution failures using autonomous error

				handling and unblocking strategies LO9: Differentiate established agentic frameworks based on their core characteristics and use cases
		Lab 2.5 Responsible AI Lab	Lab focusing on implementing guardrails in agents, demonstrating how to implement responsible AI safeguards.	LO5: Apply function calling and tool selection techniques LO10: Discuss safety risks specific to autonomous agent systems including compounding errors and unbounded solution spaces

Hands-on Activities module



SageMaker Hands-on lab

Objective: Practice using SageMaker JupyterLab to practice Agentic AI concepts learned implementing Bedrock Agentic features.

Lab 2.1: Compare Agents vs No-Agents Lab

Lab 2.2: LLM reasoning and planning with memory Lab

Lab 2.5: AI agents with guardrails



Quick Suite Hands-on lab

Objective: Practice using Amazon Quick Research feature to help create a research plan to support faculty research and study activities.

Lab 2.3: Creating a Research plan



Kiro Hands-on lab

Objective: Practice using Kiro IDE to explore how agentic AI coding assistants can help implement Bedrock Agentic features.

Lab 2.4: Creating a simple faculty's AI chat assistant

Alignment with EEP Goals

- This lesson aligns with the MLU Educator Enablement Program's strategic goals:
- **Faculty enablement:** Designed specifically for educators who will teach agentic AI concepts
- **Content adoption:** Provides faculty with foundational knowledge to confidently integrate agentic AI into their curriculum
- **Modular approach:** Part of a multi-lesson sequence that can be taught independently or as a complete module
- **Scalable education:** Prepares faculty to teach hundreds of students, multiplying the impact of the training

- **Industry-relevant skills:** Ensures students learn current, in-demand AI capabilities that prepare them for the workforce

References by session

Session 1 Resources: Reasoning and Planning Techniques

Chain-of-Thought (CoT) Reasoning Resources



code-based faculty

[Chain-of-Thought Prompting Elicits Reasoning in Large Language Models](#)

[Towards Understanding Chain-of-Thought Prompting: An Empirical Study of What Matters](#)

[Understanding Chain-of-Thought in LLMs through Information Theory](#)

[Towards Data Science's comprehensive tutorial on Chain-of-Thought prompting techniques](#)

[The Hugging Face Chain-of-Thought Leaderboard](#)

[The OpenAI Developer Cookbook](#)



code-free faculty

[Learning to reason with LLMs](#)

ReAct (Reasoning and Acting) Methodology Resources



code-based faculty

[ReAct: Synergizing Reasoning and Acting in Language Models](#)

Synergizing Reasoning and Acting in Language Model

Multi-Step Planning and Reasoning Improves Acting in LLM Agents

Language Agent Tree Search Unifies Reasoning Acting and Planning in Language Models